6. Write short notes (attempt any **three**): 3×5=15

(i) High-altitude adaptations in various vertebrate classes.

(ii) Structure and digestion in different compartments of the rumen.

(iii) Modes of reproduction in various vertebrates.

(iv) Physiological responses to desert conditions.

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[This question paper contains 4 printed pages]

Your Roll No. : 16.05.2026(M)

Sl. No. of Q. Paper : **107** I

Unique Paper Code : 2232014801

Name of the Paper : DSC-20 : Comparative Physicology of Vertebrates (NEP-UGCF)

Name of the Course : **B.Sc.(H) Zoology/B.Sc. (P) Life Science**

Semester : **VIII**

Time : 2 Hours ***Maximum Marks : 60***

Instructions for Candidates :

(a) Write your Roll No. on the top immediately on receipt of this question paper.

(b) Attempt **four** questions in all.

(c) Draw neat labelled diagrams wherever necessary.

1. (a) Define the following terms (attempt any **five**): 5×1=5

(i) Thermogenin

(ii) Hypobaric hypoxia

(iii) Hardware stomach

(iv) Root-effect hemoglobin.

(v) Ovoviviparity

(vi) Starvation

(b) Differentiate between the following (attempt any **four**) : 4×2=8

(i) Starvation and Fasting

(ii) Osmoconformers and Osmoregulators

(iii) Gill and Parabronchial respiration

(iv) Autoenzymatic and Alloenzymatic Digestion

(v) Estrous and Menstrual Cycle

(c) Expand the following (attempt any **two**) : 2×1=2

(i) HVR

(ii) BAT

(iii) HPG

2. (a) Compare the mechanisms of thermoregulation in homeotherms and poikilotherms. Add a note on the role of various hormones and ion channels therein. 10

(b) Describe the mechanisms of thermal regulation in desert animals. 5

3. (a) Delineate the fine structure of the fish gill and the bird lung, and compare the functional similarity between them. 8

(b) Examine the process of counter-current exchange of O_2 and CO_2 in fish and birds. 7

4. (a) Discuss the challenges and physiological adaptations necessary for air and aquatic respiration. 8

(b) Discuss the gradual shift in physiological fuels in response to chronic starvation in various vertebrates. 7

5. (a) Explain the general mechanisms of Osmoregulation in animals. 5

(b) Compare and contrast the challenges of osmoregulation among freshwater and marine vertebrates. 10